

Task1: Microservices Basics

Tags

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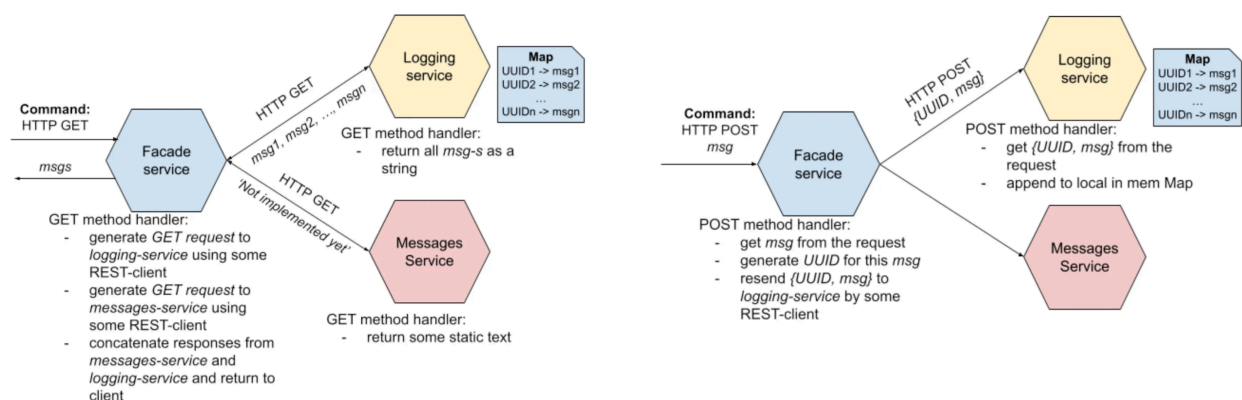
GitHub: <https://github.com/mariiaivanchenko/Software-Architecture-Task1.git>

У цьому завданні було необхідно реалізувати 3 мікросервіси, які взаємодіють між собою на основі HTTP протоколу. Головні завдання сервісів такі:

- **facade-service** - приймає POST/GET запити від клієнта
- **logging-service** - зберігає у пам'яті всі повідомлення які йому надходять, та може повертати їх
- **messages-service** - поки виступає у ролі заглушки, при зверненні до нього повертає статичне повідомлення

Базовий функціонал системи

Щоб реалізувати базовий функціонал потрібно реалізувати два методи: **GET** та **POST**. У ролі клієнта я використовувала **curl** запити.



Спершу потрібно запустити усі сервіси. Виконує це за допомогою наступних команд у різних терміналах.

```
flask --app facade-service run
flask --app logging-service run --port=5001
flask --app messages-service run --port=5002
```

POST

Коли на **facade-service** надходить **POST** запит з текстовим повідомленням, він генерує для нього унікальний UUID. Утворену пару **{UUID: message}** він надсилає за допомогою **POST** запита до **logging-service**, де той зберігає його у локальний словник та виводить у свою консоль отримане повідомлення, **facade-service** нічого не повертає.

Надсилаємо **POST** запит на **facade-service**:

```
curl -X POST -H "Content-Type: text/plain" -d msg http://127.0.0.1:5000/facade-s
```

facade-service нічого не повернув:

```
.venv> micro_basics curl -X POST -H "Content-Type: text/plain" -d msg http://127.0.0.1:5000/facade-service
.venv> micro_basics
```

Відповідні запити пройшли, успішно виконаний POST на **facade-service**, успішно виконаний POST на **logging-service** та виведено отримане повідомлення.

```
.venv> micro_basics flask --app facade-service run
* Serving Flask app 'facade-service'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
127.0.0.1 - - [16/Feb/2025 12:44:15] "POST /facade-service HTTP/1.1" 200 -
```

```
.venv> micro_basics flask --app logging-service run --port=5001
* Serving Flask app 'logging-service'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5001
Press CTRL+C to quit
Recieved messages: msg
127.0.0.1 - - [16/Feb/2025 12:44:15] "POST /logging-service HTTP/1.1" 200 -
```

Також, варіант коли надіслали кілька повідомлень:

```
● .venv➔ micro_basics curl -X POST -H "Content-Type: text/plain" -d msg http://127.0.0.1:5000/facade-service
● .venv➔ micro_basics curl -X POST -H "Content-Type: text/plain" -d msg1 http://127.0.0.1:5000/facade-service
● .venv➔ micro_basics curl -X POST -H "Content-Type: text/plain" -d msg2 http://127.0.0.1:5000/facade-service
● .venv➔ micro_basics curl http://127.0.0.1:5000/facade-service
Logging response: msg, msg1, msg2
Messages response: Messages GET is not implemented yet.
○ .venv➔ micro_basics
```

```
.venv➔ micro_basics flask --app facade-service run
* Serving Flask app 'facade-service'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
127.0.0.1 - - [16/Feb/2025 13:55:28] "POST /facade-service HTTP/1.1" 200 -
127.0.0.1 - - [16/Feb/2025 13:55:39] "POST /facade-service HTTP/1.1" 200 -
127.0.0.1 - - [16/Feb/2025 13:55:45] "POST /facade-service HTTP/1.1" 200 -
127.0.0.1 - - [16/Feb/2025 13:55:50] "GET /facade-service HTTP/1.1" 200 -
□
```

```
.venv➔ micro_basics flask --app logging-service run --port=5001
* Serving Flask app 'logging-service'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5001
Press CTRL+C to quit
Recieved messages: msg
127.0.0.1 - - [16/Feb/2025 13:55:28] "POST /logging-service HTTP/1.1" 200 -
Recieved messages: msg1
127.0.0.1 - - [16/Feb/2025 13:55:39] "POST /logging-service HTTP/1.1" 200 -
Recieved messages: msg2
127.0.0.1 - - [16/Feb/2025 13:55:45] "POST /logging-service HTTP/1.1" 200 -
127.0.0.1 - - [16/Feb/2025 13:55:50] "GET /logging-service HTTP/1.1" 200 -
□
```

GET

Коли на **facade-service** надходить **GET** запит, він генерує 2 відповідних **GET** запити до **logging-service** та **messages-service**. Після отримання **GET** запиту **logging-service** повертає усі збережені локально повідомлення у вигляді стрічки, **messages-service** повертає текст з повідомлення про неповну імплементацію. Повернувшись на **facade-service** поєднає відповіді обох сервісів та повертає клієнту.

Надсилаємо **GET** запит на **facade-service**:

```
curl http://127.0.0.1:5000/facade-service
```

facade-service повернув збережені повідомлення, яке ми надіслали раніше:

```
.venv➔ micro_basics curl -X POST -H "Content-Type: text/plain" -d msg http://127.0.0.1:5000/facade-service
.venv➔ micro_basics curl http://127.0.0.1:5000/facade-service
Logging response: msg
Messages response: Messages GET is not implemented yet.
.venv➔ micro_basics
```

Успішні запити на усіх трьох сервісах:

```
.venv➔ micro_basics flask --app facade-service run
* Serving Flask app 'facade-service'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
127.0.0.1 - - [16/Feb/2025 12:44:15] "POST /facade-service HTTP/1.1" 200 -
127.0.0.1 - - [16/Feb/2025 13:29:12] "GET /facade-service HTTP/1.1" 200 -
[]
```

```
.venv➔ micro_basics flask --app logging-service run --port=5001
* Serving Flask app 'logging-service'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5001
Press CTRL+C to quit
Received messages: msg
127.0.0.1 - - [16/Feb/2025 12:44:15] "POST /logging-service HTTP/1.1" 200 -
127.0.0.1 - - [16/Feb/2025 13:29:12] "GET /logging-service HTTP/1.1" 200 -
[]
```

```
.venv➔ micro_basics flask --app messages-service run --port=5002
* Serving Flask app 'messages-service'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5002
Press CTRL+C to quit
127.0.0.1 - - [16/Feb/2025 13:29:12] "GET /messages-service HTTP/1.1" 200 -
[]
```

Також, варіант **GET** запиту, коли не було надіслано жодного повідомлення:

```
.venv➔ micro_basics curl http://127.0.0.1:5000/facade-service
Logging response:
Messages response: Messages GET is not implemented yet.
.venv➔ micro_basics
```

Додатковий функціонал системи

Щоб реалізувати додатковий функціонал необхідно виконати:

- **retry** - у випадку проблем з надсиланням запитів, повторити кілька разів
- **deduplication** - забезпечити, що одне і те саме повідомлення не надсилалось двічі
- **gRPC протокол** - реалізувати взаємодію між **facade-service** та **logging-service** за допомогою протоколу gRPC

RETRY

Цю логіку я додала до **facade-service** для обох запитів. Задля перевірки симулюю ситуації:

- коли перші 2 запити неуспішні, а 3 успішний
- коли всі 3 неуспішні

```
# correct logging-service url
LOGGING_URL = "http://127.0.0.1:5001/logging-service"
# incorrect logging-service url to test RETRY
LOGGING_URL = "http://127.0.0.1:9999"
```

• POST

1. Надіслали **POST** запит, повідомлення про помилку нема, запит був виконаний успішно:

```
.venv> micro_basics curl -X POST -H "Content-Type: text/plain" -d msg http://127.0.0.1:5000/facade-service
.venv> micro_basics
```

На **facade-service** відбувалось кілька спроб, 2 неуспішні, 1 успішна:

```
.venv> micro_basics flask --app facade-service run
* Serving Flask app 'facade-service'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
Due to delays, poor connection or no response, exception occurred: HTTPConnectionPool(host='127.0.0.1', port=9999): Max retries exceeded with url: / (Caused by NewConnectionError('<urllib3.connection.HTTPConnection object at 0x10d155a90>: Failed to establish a new connection: [Errno 61] Connection refused'))
Retrying the operation.
Due to delays, poor connection or no response, exception occurred: HTTPConnectionPool(host='127.0.0.1', port=9999): Max retries exceeded with url: / (Caused by NewConnectionError('<urllib3.connection.HTTPConnection object at 0x10d15c190>: Failed to establish a new connection: [Errno 61] Connection refused'))
Retrying the operation.
127.0.0.1 - - [16/Feb/2025 13:43:47] "POST /facade-service HTTP/1.1" 200 -
```

2. Надіслали **POST** запит, з'явилося повідомлення про помилку, запит не був виконаний, навіть після кількох спроб:

```
.venv> micro_basics curl -X POST -H "Content-Type: text/plain" -d msg http://127.0.0.1:5000/facade-service
Due to delays, poor connection or no response, exception occurred! Retrying attempts are over.
.venv> micro_basics
```

На **facade-service** відбувалось 3 неуспішні спроби:

```
.venv➔ micro_basics flask --app facade-service run
* Serving Flask app 'facade-service'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
Due to delays, poor connection or no response, exception occurred: HTTPConnectionPool(host='127.0.0.1', port=9999): Max retries exceeded with url: / (Caused by NewConnectionError('<urllib3.connection.HTTPConnection object at 0x109675a90>: Failed to establish a new connection: [Errno 61] Connection refused'))
Retrying the operation.
Due to delays, poor connection or no response, exception occurred: HTTPConnectionPool(host='127.0.0.1', port=9999): Max retries exceeded with url: / (Caused by NewConnectionError('<urllib3.connection.HTTPConnection object at 0x10967c190>: Failed to establish a new connection: [Errno 61] Connection refused'))
Retrying the operation.
Due to delays, poor connection or no response, exception occurred: HTTPConnectionPool(host='127.0.0.1', port=9999): Max retries exceeded with url: / (Caused by NewConnectionError('<urllib3.connection.HTTPConnection object at 0x10967ca50>: Failed to establish a new connection: [Errno 61] Connection refused'))
Retrying the operation.
127.0.0.1 -- [16/Feb/2025 13:49:00] "POST /facade-service HTTP/1.1" 500 -
```

• GET

1. Надіслали **GET** запит, повідомлення про помилку нема, запит був виконаний успішно:

```
.venv➔ micro_basics curl http://127.0.0.1:5000/facade-service
Logging response: msg
Messages response: Messages GET is not implemented yet.
.venv➔ micro_basics
```

На **facade-service** відбувалось кілька спроб, 2 неуспішні, 1 успішна:

```
.venv➔ micro_basics flask --app facade-service run
* Serving Flask app 'facade-service'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
Due to delays, poor connection or no response, exception occurred: HTTPConnectionPool(host='127.0.0.1', port=9999): Max retries exceeded with url: / (Caused by NewConnectionError('<urllib3.connection.HTTPConnection object at 0x1119b5a90>: Failed to establish a new connection: [Errno 61] Connection refused'))
Retrying the operation.
Due to delays, poor connection or no response, exception occurred: HTTPConnectionPool(host='127.0.0.1', port=9999): Max retries exceeded with url: / (Caused by NewConnectionError('<urllib3.connection.HTTPConnection object at 0x1119bc190>: Failed to establish a new connection: [Errno 61] Connection refused'))
Retrying the operation.
127.0.0.1 -- [16/Feb/2025 13:51:46] "GET /facade-service HTTP/1.1" 200 -
```

2. Надіслали **POST** запит, з'явилося повідомлення про помилку, запит не був виконаний, навіть після кількох спроб:

```
.venv➔ micro_basics curl http://127.0.0.1:5000/facade-service
Due to delays, poor connection or no response, exception occurred! Retrying attempts are over.
.venv➔ micro_basics
```

На **facade-service** відбувалось 3 неуспішні спроби:

```
.venv> micro_basics flask --app facade-service run
* Serving Flask app 'facade-service'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
Due to delays, poor connection or no response, exception occurred: HTTPConnectionPool(host='127.0.0.1', port=9999): Max retries exceeded with url: / (Caused by NewConnectionError('<urllib3.connection.HTTPConnection object at 0x10c0d5a90>: Failed to establish a new connection: [Errno 61] Connection refused'))
Retrying the operation.
Due to delays, poor connection or no response, exception occurred: HTTPConnectionPool(host='127.0.0.1', port=9999): Max retries exceeded with url: / (Caused by NewConnectionError('<urllib3.connection.HTTPConnection object at 0x10c0dc190>: Failed to establish a new connection: [Errno 61] Connection refused'))
Retrying the operation.
Due to delays, poor connection or no response, exception occurred: HTTPConnectionPool(host='127.0.0.1', port=9999): Max retries exceeded with url: / (Caused by NewConnectionError('<urllib3.connection.HTTPConnection object at 0x10c0dca50>: Failed to establish a new connection: [Errno 61] Connection refused'))
Retrying the operation.
127.0.0.1 - - [16/Feb/2025 13:53:19] "GET /facade-service HTTP/1.1" 500 -
█
```

DEDUPLICATION

Ця логіка має забезпечувати, щоб однакові повідомлення не повторювались навіть про декількох спробах доступу до сервісу. Для тесту буде відбуватись 2 неуспішні спроби та 1 успішна.

Можна побачити, що однакових повідомлень не було збережено:

```
.venv> micro_basics curl -X POST -H "Content-Type: text/plain" -d msg http://127.0.0.1:5000/facade-service
.venv> micro_basics curl http://127.0.0.1:5000/facade-service
Logging response: msg
Messages response: Messages GET is not implemented yet.
.venv> micro_basics curl -X POST -H "Content-Type: text/plain" -d msg http://127.0.0.1:5000/facade-service
.venv> micro_basics curl http://127.0.0.1:5000/facade-service
Logging response: msg
Messages response: Messages GET is not implemented yet.
.venv> micro_basics curl -X POST -H "Content-Type: text/plain" -d msg2 http://127.0.0.1:5000/facade-service
.venv> micro_basics curl http://127.0.0.1:5000/facade-service
Logging response: msg, msg2
Messages response: Messages GET is not implemented yet.
.venv> micro_basics █
```

На **facade-service** відбувалось 3 **POST**, з кількома спробами, та 3 **GET** запити:

```
.venv➤ micro_basics flask --app facade-service run
* Serving Flask app 'facade-service'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
Due to delays, poor connection or no response, exception occurred: HTTPConnectionPool(host='127.0.0.1', port=9999): Max retries exceeded with url: / (Caused by NewConnectionError('<urllib3.connection.HTTPConnection object at 0x101da9a90>: Failed to establish a new connection: [Errno 61] Connection refused'))
Retrying the operation.
Due to delays, poor connection or no response, exception occurred: HTTPConnectionPool(host='127.0.0.1', port=9999): Max retries exceeded with url: / (Caused by NewConnectionError('<urllib3.connection.HTTPConnection object at 0x101db0190>: Failed to establish a new connection: [Errno 61] Connection refused'))
Retrying the operation.
127.0.0.1 - - [16/Feb/2025 14:01:50] "POST /facade-service HTTP/1.1" 200 -
127.0.0.1 - - [16/Feb/2025 14:02:07] "GET /facade-service HTTP/1.1" 200 -
Due to delays, poor connection or no response, exception occurred: HTTPConnectionPool(host='127.0.0.1', port=9999): Max retries exceeded with url: / (Caused by NewConnectionError('<urllib3.connection.HTTPConnection object at 0x101d92210>: Failed to establish a new connection: [Errno 61] Connection refused'))
Retrying the operation.
Due to delays, poor connection or no response, exception occurred: HTTPConnectionPool(host='127.0.0.1', port=9999): Max retries exceeded with url: / (Caused by NewConnectionError('<urllib3.connection.HTTPConnection object at 0x101d47df0>: Failed to establish a new connection: [Errno 61] Connection refused'))
Retrying the operation.
127.0.0.1 - - [16/Feb/2025 14:02:09] "POST /facade-service HTTP/1.1" 200 -
127.0.0.1 - - [16/Feb/2025 14:02:11] "GET /facade-service HTTP/1.1" 200 -
Due to delays, poor connection or no response, exception occurred: HTTPConnectionPool(host='127.0.0.1', port=9999): Max retries exceeded with url: / (Caused by NewConnectionError('<urllib3.connection.HTTPConnection object at 0x101dcc050>: Failed to establish a new connection: [Errno 61] Connection refused'))
Retrying the operation.
Due to delays, poor connection or no response, exception occurred: HTTPConnectionPool(host='127.0.0.1', port=9999): Max retries exceeded with url: / (Caused by NewConnectionError('<urllib3.connection.HTTPConnection object at 0x101dcc8d0>: Failed to establish a new connection: [Errno 61] Connection refused'))
Retrying the operation.
127.0.0.1 - - [16/Feb/2025 14:02:19] "POST /facade-service HTTP/1.1" 200 -
127.0.0.1 - - [16/Feb/2025 14:02:21] "GET /facade-service HTTP/1.1" 200 -

```

На **logging-service** відбувалось 3 **POST** та 3 **GET** запити:

```
.venv➤ micro_basics flask --app logging-service run --port=5001
* Serving Flask app 'logging-service'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5001
Press CTRL+C to quit
Received messages: msg
127.0.0.1 - - [16/Feb/2025 14:05:35] "POST /logging-service HTTP/1.1" 200 -
127.0.0.1 - - [16/Feb/2025 14:05:38] "GET /logging-service HTTP/1.1" 200 -
127.0.0.1 - - [16/Feb/2025 14:05:40] "POST /logging-service HTTP/1.1" 200 -
127.0.0.1 - - [16/Feb/2025 14:05:41] "GET /logging-service HTTP/1.1" 200 -
Received messages: msg2
127.0.0.1 - - [16/Feb/2025 14:05:45] "POST /logging-service HTTP/1.1" 200 -
127.0.0.1 - - [16/Feb/2025 14:05:46] "GET /logging-service HTTP/1.1" 200 -

```

gRPC ПРОТОКОЛ

Щоб виконувати ту саму комунікацію між **facade-service** та **logging-service** за допомогою протоколу gRPC необхідно виконати кілька етапів:

- встановити необхідну бібліотеку

```
pip install grpcio grpcio-tools
```

- спершу написати відповідний **.proto** файл, який буде містити опис того, що буде відбуватись та з якими даними

- потім за допомогою команди згенерувати grpc файли

```
python -m grpc_tools.protoc -I. --python_out=. --grpc_python_out=. micro_b
```

- імплементувати сервера та клієнта, у моєму випадку **facade-service** (клієнт) та **logging-service** (сервер)

Для запуску програм такі команди:

```
python3 facade-service-grpc.py
python3 logging-service-grpc.py
```

```
.venv➔ micro_basics python3 facade-service-grpc.py
* Serving Flask app 'facade-service-grpc'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 112-644-220
```

```
.venv➔ micro_basics python3 logging-service-grpc.py
Logging Service gRPC server running on port 5001...
```

Запити:

```
curl http://127.0.0.1:5000/facade-service-grpc
curl -X POST -H "Content-Type: text/plain" -d "Hello from facade" 127.0.0.1:5000
```

Приклад надсилання запитів:

```
● .venv➔ micro_basics curl http://127.0.0.1:5000/facade-service-grpc
  Logging response:
● .venv➔ micro_basics curl -X POST -H "Content-Type: text/plain" -d "Hello from facade" 127.0.0.1:5000/facade-service-grpc
  Hello from facade
● .venv➔ micro_basics curl http://127.0.0.1:5000/facade-service-grpc
  Logging response: Hello from facade
○ .venv➔ micro_basics
```

```
.venv➔ micro_basics python3 facade-service-grpc.py
* Serving Flask app 'facade-service-grpc'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 112-644-220
127.0.0.1 - - [16/Feb/2025 14:15:19] "GET /facade-service-grpc HTTP/1.1" 200 -
127.0.0.1 - - [16/Feb/2025 14:15:24] "POST /facade-service-grpc HTTP/1.1" 200 -
127.0.0.1 - - [16/Feb/2025 14:15:26] "GET /facade-service-grpc HTTP/1.1" 200 -
█
```